

ActionPath

Crisis-to-Action Translator — Enhanced PRD v2.0

USAII Global AI Hackathon 2026 · Challenge 1A · High School Track

AI-first executive function tool for ADHD students — powered entirely by Gemini AI

Version	Track	AI Stack	Build Time
v2.0 — Enhanced	High School 1A	Gemini 2.5 Pro / Flash / Flash-Lite	6 Days

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1. Problem Statement & User Research

Schools produce a relentless stream of communications — emails, LMS posts, newsletters, PDF forms, Google Classroom announcements. Inside them are deadlines, permission slips, scholarship links, and required actions. For most students, filtering signal from noise is merely annoying. For ADHD students with executive dysfunction, it is cognitively near-impossible — a documented neurological barrier, not a character flaw.

"Every week, an ADHD student opens a school email and misses something critical — not because they don't care, but because finding the action item inside 500 words of text is a task their brain makes nearly impossible."

The Failure Cascade

Step	What Happens	Why It Fails for ADHD
1	Student receives 400-word school email	Inbox is already overwhelming; email is deprioritized
2	Email contains 3 buried action items	Dense prose = impossible signal extraction
3	Student opens email, feels overwhelmed	Executive dysfunction blocks task initiation
4	Email is closed or left unread	Out of sight = out of working memory entirely
5	Deadline passes	No external system flagged the consequence
6	Grade impact / missed opportunity	Systemic failure blamed on individual student

Primary User Profile

Diagnosis	ADHD — inattentive, hyperactive, or combined type (DSM-5)
Core difficulty	Executive dysfunction: task initiation, noise filtering, delayed-action follow-through
Grade	High school, grades 9–12

Communication medium	School email, Google Classroom, Canvas LMS, Schoology, PDF newsletters
Tech comfort	High — uses phone and laptop daily, comfortable with web apps and OAuth flows
Support system	May have IEP/504 plan; no tool exists specifically for communication processing
Primary pain moment	Opening a long school email and not knowing what to do next
Secondary pain moment	Forgetting about a deadline that was buried in a message two weeks ago
Emotional state	Anxious, avoidant, ashamed — not lazy or careless

Secondary Users

School counselors and special education coordinators who support ADHD students benefit indirectly. A read-only counselor view means students arrive to appointments having already processed their action items — reducing the time spent on logistics and increasing time for actual support. Parents using the family view get a window into their child's academic obligations without needing to see the raw communications.

2. AI Architecture — 7-Stage Gemini Pipeline

ActionPath v2 replaces the original 5-stage Claude pipeline with a fully AI-powered 7-stage Gemini pipeline. Every single stage — including the pre-processing classifier — makes a Gemini API call. No rule-based fallbacks. No heuristics masquerading as AI. Each agent has one focused job, a defined JSON output schema, and is assigned to the most cost-efficient Gemini model for that task.

STAGE 1

INTAKE CLASSIFIER AGENT

■ gemini-3.1-flash-lite

Receives raw input (pasted text, .txt, .pdf, .eml, .html). Classifies communication type: `deadline_notice` / `general_update` / `emergency` / `form_request` / `payment_required` / `event_invitation` / `policy_change`. Estimates `cognitive_load_score` (0–10) based on word count, sentence complexity, action density, and urgency language. Chooses the optimal downstream prompt template. Returns a routing decision that controls all subsequent agents.

Output schema:

```
{ "comm_type": "deadline_notice",  
  "cognitive_load_score": 7.4,  
  "word_count": 412,  
  "action_density": 0.08,  
  "routing_template": "DEADLINE_V2",  
  "language_complexity": "high" }
```

STAGE 2

ENTITY & CONTEXT EXTRACTION AGENT

■ gemini-3.1-pro-preview

Full deep extraction using Gemini 2.5 Pro — the most capable model, used here because extraction accuracy is critical and errors cascade to all downstream stages. Extracts: every task/deadline/action item, the exact source sentence, people mentioned, amounts/fees, URLs, form names, room numbers, and subject associations. Outputs structured JSON with span-level source linking.

Output schema:

```
"items": [{ "task": "...", "deadline": "2026-06-17",  
  "source_sentence": "exact quote...",  
  "item_type": "form|deadline|payment|meeting|action",  
  "people": ["Ms. Rivera"], "url": null,  
  "subject": "Biology", "confidence": 0.94 }]
```

STAGE 3

SCORING & COLLISION AGENT

■ gemini-3.5-flash

Takes extraction output. Scores each item on four axes: urgency (days until deadline, 1–10), effort (low = click / medium = fill form / high = write / very_high = multi-step project), dependency (does this block or require another item), and consequence_severity (what happens if missed: grade / opportunity / social / financial). Detects deadline collisions (3+ tasks on same day) and suggests reordering. Outputs a ranked priority list.

Output schema:

```
"scored_items": [{ "task_id": "t1",
"urgency": 9, "effort": "low",
"consequence": "grade", "dependency": null,
"composite_score": 8.7 }],
"collision_days": ["2026-06-17"],
"reorder_suggestion": "Move lab fee to today"
```

STAGE 4

CONTEXT FUSION AGENT

■ gemini-3.1-flash-lite

Merges the scored items with the student's stored context: completed tasks, known subjects, past deadline patterns, and preferred working times. Deduplicates items that match previously seen tasks. Flags items that are recurring (weekly newsletter format detected). Lightweight — uses Flash-Lite because the reasoning is straightforward merge-and-deduplicate logic, not creative reasoning.

Output schema:

```
"merged_items": [...],
"deduplicated": ["t3 matches seen on 2026-06-10"],
"recurring_detected": true,
"student_subject_match": { "Biology": ["t1", "t4"] }
```

STAGE 5

DAY-BY-DAY PLANNING AGENT

■ gemini-3.5-flash

Takes merged scored items. Generates a concrete daily action plan across three horizons: Today / Tomorrow / This Week / Later. Applies cognitive load balancing — never assigns more than 3 high-effort tasks to a single day. If a day is overloaded, suggests which items to move and why. Generates an estimated time budget per day (e.g. 'Today: ~45 minutes total'). Accounts for the student's stored 'busy days' preference.

Output schema:

```
"plan": {
  "today": [{"task_id": "t1", "est_mins": 5}],
  "tomorrow": [{"task_id": "t2", "est_mins": 20}],
  "this_week": [...], "later": [...] },
  "load_warning": "Wednesday has 4 items (42 min total)",
  "daily_budgets": {"today": 25, "tomorrow": 20}
```

STAGE 6

LANGUAGE REWRITER AGENT

■ gemini-3.5-flash

Rewrites every task description into plain, calm, ADHD-optimized language. Rules: max 12 words per task, active voice, no bureaucratic phrasing, no passive constructions, start with a verb. Adds a one-sentence 'why this matters' if the task has a grade/opportunity consequence. Also generates a 'start cue' — the single first physical action the student should take (e.g. 'Open Google Forms → ...'). This removes the initiation barrier.

Output schema:

```
"rewritten": "Sign field trip form. Hand to room 204.",
"why_it_matters": "Missing this means you cannot attend.",
"start_cue": "Open your backpack and find the blue form.",
"original": "Please return the signed permission slip..."
```

STAGE 7

QUALITY ASSURANCE AGENT

■ gemini-3.1-flash-lite

Final pass QA agent — new in v2. Checks the full pipeline output for: missing deadlines (compares extracted count vs. a re-scan), confidence < 0.7 (flags for student review), any task with no start_cue (requests regeneration), overloaded days not flagged with warnings, and any rewritten task over 12 words. Returns a qa_passed boolean and a list of qa_issues for the UI to surface.

Output schema:

```
"qa_passed": false,
"qa_issues": [
  { "task_id": "t3", "issue": "low_confidence",
    "action": "flag_for_review" },
  { "task_id": "t5", "issue": "rewrite_too_long" }
]
```

3. Gemini Model Assignment Strategy

Every Gemini model has a different cost-accuracy-latency tradeoff. ActionPath assigns models deliberately — the most capable model only where accuracy is critical, the most efficient model everywhere else. This keeps the pipeline fast and within free-tier limits during the hackathon.

Model	Stages Used	Why This Model	Strengths
gemini-3.1-pro-preview	Stage 2 (Extraction)	Extraction errors cascade to all downstream stages — use the best model here	Highest accuracy, best at span-level extraction and nuanced context
gemini-3.5-flash	Stages 3, 5, 6 (Scoring, Planning, Rewriting)	Medium complexity reasoning — needs creativity + speed, not max accuracy	Excellent reasoning, fast, cost-efficient for multi-step logic
gemini-3.1-flash-lite	Stages 1, 4, 7 (Classify, Merge, QA)	Simple classification / merge / check tasks — fastest and cheapest is correct	Very low latency, minimal cost, sufficient for structured tasks

API Call Pattern: All calls use Gemini's JSON mode (response_mime_type: 'application/json') to guarantee structured output. Each stage passes its JSON output directly as the next stage's input — no string parsing, no regex. The pipeline is fully typed end-to-end.

4. Complete Feature Set

4.1 Core MVP Features

Feature	Description	AI Stage
Multi-format input	Paste raw text OR upload .pdf / .txt / .eml / .html. Drag-and-drop supported.	Pre-pipeline
7-stage AI pipeline	Full Gemini pipeline: classify → extract → score → fuse → plan → rewrite → QA	Stages 1–7
Interactive checklist	Tasks as checkboxes. Check = done. Persists across sessions via localStorage.	Frontend
Week view timeline	7-day visual timeline. Collision days highlighted amber. Overloaded days shown in red.	Frontend
Source sentence tooltip	Hover/tap any task: see exact sentence it was extracted from. Confidence badge shown.	Stage 2
Cognitive load warning	3+ tasks on one day triggers a warning + suggested reordering.	Stage 3 + 5
Start cue cards	Each task shows the single first physical action to eliminate initiation barrier.	Stage 6
Calendar export (.ics)	One-click export. Works with Google Calendar, Apple Calendar, Outlook.	Frontend
Plain language toggle	Switch between ADHD-optimized rewrite and original extracted text.	Stage 6
QA confidence badges	High / Medium / Review badge on each task. Review items flagged in red.	Stage 7
Why-it-matters label	Consequence shown beneath tasks with grade/opportunity impact.	Stage 6
Time budget display	Daily estimated time total (e.g. 'Today: ~35 min'). Per-task effort shown.	Stage 5

Feature	Description	AI Stage
Edit/delete any task	Student can correct, delete, or add tasks manually. AI output is a starting point.	Frontend

4.2 Stretch Features (Vibe-Code Bonus Round)

These features are designed for rapid vibe-coding. Each one is self-contained, adds real value, and can be implemented independently. Tackle in any order.

Feature	What It Does	Build Complexity
Voice input mode	Record a verbal description of your assignment. Gemini Flash transcribes + processes.	Low — Web Speech API + existing pipeline
Google Classroom OAuth	Pull messages directly. Student connects account; no copy-paste needed.	Medium — OAuth flow + Classroom API
Mood check-in	Before showing the plan, ask: 'How are you feeling? (1–5)'. Adjusts day's task count accordingly.	Low — one extra UI step + conditional logic
Focus mode (Pomodoro)	Start any task → 25-min timer appears. Completion auto-checks the task.	Low — timer component + localStorage
Streak tracker	Counts consecutive days the student completed their Today list. Shows streak badge.	Low — localStorage date comparison
Email forwarding address	Student gets a unique address (e.g. student@actionpath.app) to forward emails directly.	High — requires email receiving service
Parent/counselor view	Read-only shareable link to current action plan. No login required for viewer.	Medium — unique URL + read-only state
Recurring detection	Detects repeated newsletter formats and skips low-value boilerplate automatically.	Medium — Stage 4 enhancement
Effort feedback loop	After completing a task: 'How long did this actually take?' Improves future scoring.	Low — feedback modal + localStorage

Feature	What It Does	Build Complexity
Dark mode	Full dark mode toggle. Persists via localStorage.	Low — Tailwind dark: classes
Subject color coding	Each subject (Biology, English, etc.) gets a color. Tasks inherit their subject color.	Low — subject → color map
Smart reminders via push	Browser push notifications for tasks due in the next 24 hours.	Medium — Service Worker + Notification API
Multi-communication batch	Process 3 emails at once. Pipeline runs in parallel. Results merged and deduplicated.	Medium — Promise.all() + Stage 4 enhancement
Accessibility mode	High-contrast mode + screen-reader labels + keyboard navigation for all interactions.	Low — ARIA labels + Tailwind

5. Responsible AI Design

Responsible AI is not an afterthought in ActionPath — it is structural. Every element below is built into the product UI, not just described in documentation.

Element	Risk Addressed	Mitigation Built Into Product
Source sentence linking	AI extracts a deadline incorrectly → student misses it or acts on wrong date	Every task shows the exact source sentence. Student can verify before acting.
Confidence badges	AI is uncertain but presents output with false confidence	High / Medium / Review badge on every item. Review items shown in red with edit prompt.
Human-in-the-loop by design	Student over-relies on AI and stops reading communications	AI never submits, signs, pays, or confirms. Every action requires student input. Checkboxes are not auto-checked.
Original always accessible	Student loses context of the original communication	'View original' link always visible beneath every processed communication.
Edit + delete any item	AI output is treated as ground truth	Every task is editable and deletable. AI output is a starting point, not final truth.
QA agent (Stage 7)	Pipeline produces low-quality output that student can't detect	Stage 7 re-scans output and flags issues. QA failures shown in UI before student sees results.
Privacy by design	Sensitive school communications stored on servers indefinitely	No communication content stored on any server. localStorage holds only task completion state. Per-request processing only.
Onboarding disclaimer	Student doesn't understand AI limitations before using tool	First-use onboarding screen explains: 'ActionPath helps you find action items. Always verify important deadlines directly.'
Counselor oversight view	AI replaces human support system entirely	Read-only counselor link enables trusted adult oversight. Tool is positioned as augmentation, not replacement.

Element	Risk Addressed	Mitigation Built Into Product
No autonomous actions	AI acts on behalf of student without consent	Zero autonomous actions. AI recommends sequence and wording. Student makes every decision.

6. Data Schema & JSON Contracts

The complete JSON contract for the full pipeline output. Each stage's output is a subset of this master schema.

```
{
  "pipeline_version": "2.0",
  "processed_at": "2026-06-15T10:30:00Z",
  "stage_1_classify": {
    "comm_type": "deadline_notice",
    "cognitive_load_score": 7.4,
    "word_count": 412,
    "action_density": 0.08,
    "routing_template": "DEADLINE_V2",
    "model": "gemini-3.1-flash-lite"
  },
  "stage_2_extract": {
    "items": [
      {
        "task_id": "t1",
        "task": "Return signed field trip permission form",
        "deadline": "2026-06-17",
        "days_remaining": 2,
        "source_sentence": "Please return the signed form to room 204 by Wednesday.",
        "item_type": "form",
        "people": ["Ms. Rivera"],
        "subject": "Biology",
        "url": null,
        "confidence": 0.94,
        "model": "gemini-3.1-pro-preview"
      }
    ]
  },
  "stage_3_score": {
    "scored_items": [
      { "task_id": "t1", "urgency": 9, "effort": "low",
        "consequence": "social", "dependency": null,
        "composite_score": 8.7 }
    ],
    "collision_days": [],
    "model": "gemini-3.5-flash"
  },
}
```

```
"stage_5_plan": {
  "plan": {
    "today": [{"task_id": "t1", "est_mins": 5}],
    "tomorrow": [],
    "this_week": [],
    "later": []
  },
  "daily_budgets": {"today": 5},
  "load_warning": null,
  "model": "gemini-3.5-flash"
},
"stage_6_rewrite": {
  "rewritten_items": [
    {
      "task_id": "t1",
      "rewritten": "Sign field trip form. Hand to room 204.",
      "why_it_matters": "Missing this means you cannot go on the trip.",
      "start_cue": "Open your backpack and find the blue permission slip.",
      "model": "gemini-3.5-flash"
    }
  ]
},
"stage_7_qa": {
  "qa_passed": true,
  "qa_issues": [],
  "model": "gemini-3.1-flash-lite"
}
}
```

7. Tech Stack & File Structure

Layer	Tool	Why
Frontend	Next.js 14 (React)	App Router, fast deploy on Vercel, excellent for interactive UI
Styling	Tailwind CSS	Rapid styling, dark mode support, no CSS files needed
AI Pipeline	Gemini API (google-generativeai SDK)	All 7 stages. JSON mode guaranteed output.
Models	gemini-3.1-pro-preview, gemini-3.5-flash, gemini-3.1-flash-lite	Assigned per-stage by complexity
Persistence	localStorage (browser)	Completed tasks + student context. No backend needed. Privacy by design.
Calendar export	ics.js	Client-side .ics generation. One function call.
PDF parsing	pdf-parse (npm) or pdfjs-dist	Client-side PDF text extraction before sending to pipeline
Deployment	Vercel (free tier)	One command deploy. Live URL for judges instantly.
API key handling	Next.js API routes (/api/pipeline)	Never expose Gemini key to client. All AI calls server-side.
State management	React useState + useReducer	No Redux needed for this scope

Recommended File Structure

actionpath/
app/
page.tsx ← Main UI: input + results
checklist/page.tsx ← Interactive checklist view
week/page.tsx ← Week timeline view
api/

pipeline/route.ts ← Server: runs all 7 Gemini stages
export/route.ts ← Generates .ics calendar file
components/
TaskCard.tsx ← Single task with tooltip + badges
WeekView.tsx ← 7-day timeline
LoadWarning.tsx ← Cognitive load warning banner
StartCue.tsx ← Start cue card component
ConfidenceBadge.tsx ← High / Medium / Review badge
lib/
gemini.ts ← Gemini client + all 7 stage functions
schema.ts ← TypeScript types for all JSON schemas
storage.ts ← localStorage read/write helpers
ics.ts ← .ics file generator
data/
demo-emails/ ← 5 synthetic school emails for demo

8. 6-Day Build Plan

Designed for a team of 2–3 vibe-coders with Claude/Cursor/Copilot assistance. Every day has a clear deliverable. Day 6 is buffer — do not add features.

Day	Goal	Deliverable
Day 1	Architecture + Stage 1 & 2	Next.js project scaffold. Gemini client configured. Stage 1 (Classify) and Stage 2 (Extract) working with JSON output. Test on 3 synthetic emails.
Day 2	Stages 3–7 + pipeline wire-up	All 7 stages running in sequence. /api/pipeline route accepts text, returns full master JSON. Basic input → JSON output demo in browser.
Day 3	Core UI: checklist + week view	Interactive checklist with checkboxes. Week view with 7-day timeline. Source sentence tooltips. Confidence badges. Cognitive load warnings.
Day 4	Calendar export + start cues + polish	.ics export working. Start cue cards visible on each task. Plain language toggle. localStorage persistence. Mobile-responsive layout.
Day 5	Demo data + stretch features	5 synthetic demo emails created. Pick 2–3 stretch features from Section 4.2. Record demo video. Write Devpost copy.
Day 6	Buffer — fix and submit	Fix bugs found in cold testing. Polish UI. Submit to Devpost before deadline. Do not add new features.

9. Demo Script & Judging Rubric Mapping

Demo Video Script (3 minutes)

Timestamp		Content	Judging Dimension
0:00 – 0:25	–	Problem statement. Show a dense, realistic 500-word school email on screen. Don't read it — just show it. Say: 'Every week, an ADHD student opens an email like this and misses something important — not because they don't care, but because finding the action item in 500 words is cognitively impossible.'	Problem (30%)
0:25 – 0:45	–	User and pain. Name the user. Show the persona card. Describe executive dysfunction in one sentence: difficulty initiating tasks, filtering noise, and acting on delayed information.	Problem (30%)
0:45 – 1:45	–	Live demo. Paste the email into ActionPath. Show the pipeline running (7 stage progress indicator). Show the output: checklist, week view, cognitive load warning, start cue cards. Hover a task to show source sentence + confidence badge. Click Export to Calendar.	Solution (20%) + AI Reasoning (20%)
1:45 – 2:20	–	AI architecture. Show the 7-agent diagram. Name each model: Flash-Lite for classification, Pro for extraction, Flash for scoring/planning/rewriting, Flash-Lite for QA. Show the JSON output. Say: 'This is not a summary. This is a structured reasoning pipeline with 7 dedicated AI agents.'	AI Reasoning (20%)
2:20 – 2:45	–	Responsible AI. Show source sentence feature. Confidence badges. Edit/delete any task. Say: 'AI never acts. Student reviews and decides.' Mention no content stored on servers.	Responsible AI (10%)
2:45 – 3:00	–	Impact close. Before: missed deadline, grade impact, shame. After: student knows exactly what to do today, with a start cue, in under 12 words. Show the streak tracker. Close with project name and tagline.	Impact (20%)

Judging Rubric Mapping

Dimension	Weight	How ActionPath Scores Maximum
Problem understanding	30%	Named user (ADHD student), named behavior (executive dysfunction), named moment (opening school email), named consequence (missed deadline, grade impact). Judge understands the problem in under 10 seconds.
AI reasoning	20%	7 distinct Gemini agents with different models and different jobs. Each stage is explainable. Judges can see JSON output at each step. Model assignment is deliberately justified.
Solution design	20%	Paste email → interactive checklist in under 5 seconds. Live demo requires no setup. Output is a usable product with start cues, week view, and calendar export — not a text block.
Impact	20%	Before: missed deadline, grade impact, shame. After: student acts with confidence, knows exactly what to do today, sees the first physical step. Emotional and measurable impact visible in 30 seconds.
Responsible AI	10%	10 distinct responsible AI elements. Source sentence verification, confidence badges, human-in-the-loop, QA agent, privacy by design, counselor oversight view. Built in, not decorative.

10. Devpost Submission Copy

Project Description

ActionPath is an AI-powered executive function tool for ADHD high school students who struggle with deadlines buried inside school communications. Students paste or upload a school email, LMS post, or PDF. A 7-stage Gemini AI pipeline — powered entirely by Gemini 2.5 Pro, Gemini 2.5 Flash, and Gemini 2.0 Flash-Lite — classifies the communication, extracts every deadline and required action with span-level source linking, scores each item by urgency/effort/consequence, fuses with the student's context, generates a day-by-day action plan with time budgets, rewrites every task in plain language under 12 words with a start cue, and QA-checks its own output. The result is an interactive checklist with a week view, cognitive load warnings, confidence badges, start cue cards, and one-click calendar export. AI recommends. Student decides and acts.

AI Architecture (for Devpost)

Stage 1 — Intake Classifier	gemini-3.1-flash-lite	Detects communication type and cognitive load score. Routes to correct prompt template.
Stage 2 — Extraction Agent	gemini-3.1-pro-preview	Extracts all tasks, deadlines, source sentences, people, URLs with confidence scores.
Stage 3 — Scoring Agent	gemini-3.5-flash	Scores each item: urgency, effort, consequence, dependency. Detects deadline collisions.
Stage 4 — Context Fusion Agent	gemini-3.1-flash-lite	Merges scored items with student's stored history. Deduplicates recurring items.
Stage 5 — Planning Agent	gemini-3.5-flash	Generates day-by-day plan with time budgets and cognitive load balancing.
Stage 6 — Language Rewriter	gemini-3.5-flash	Rewrites every task: max 12 words, active voice, start cue, why-it-matters label.
Stage 7 — QA Agent	gemini-3.1-flash-lite	Validates pipeline output. Flags low-confidence items and overloaded days.

Tagline

"ActionPath: Seven AI agents working together so one ADHD student never misses a deadline again."